Nate Cirino

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Available May 2027

Education

Northeastern University May 2027 | MGPA: 3.6/4.0
B.S. in Computer Science, Systems Concentration *Boston, MA*

Coursework: Computer Systems, Digital Design & Computer Architecture, Computer Networks, Systems Security, Cybersecurity, Algorithms & Data Structures, Object-Oriented Design, Foundations of Software Engineering, Foundations of Data Science

Experience

Wayfair Jul 2026 – Present
Software Engineering Co-op *Boston, MA*

- Own front-of-house device stack across all Wayfair physical retail brands (AllModern, Birch Lane, Joss & Main, Perigold): e-label hardware, mobile, and customer/associate-facing POS
- Building a supplier-agnostic e-label API that abstracts 4 device suppliers behind one hardware interface, decoupling application software from vendor firmware and powering new outlet launches
- Shipped POS checkout flows: in-store scan-to-basket customer switching and live associate-entry customer sync

Wolters Kluwer Jul 2025 – Jun 2026
Software Engineer *Boston, MA*

- Overhauled Relay JMS broker serialization with json-io across 20+ microservices, eliminating timeouts at scale
- Implemented OIDC/IdP authentication via Azure across 20 enterprise health systems, securing 10K+ clinicians
- Built Java Spring microservice for CME credit issuance inside Expert AI's clinician-facing workflow
- Integrated Expert AI into CME accrual/redemption microservices, expanding GenAI CDS to 250K+ clinicians
- Optimized Expert AI search and filtering across 20+ microservices, improving clinical query relevance at scale

SculptAI Jan 2025 – Aug 2026
Software Architect *New York, NY*

- Maintained 99.99% uptime across 1-2M monthly model calls for 2K paying users via AWS WAF, Datadog, and K8s
- Architected Flask backend with RAG pipeline for personalized workout recommendations using Bedrock KB
- Led eng org restructure at $1M-funded startup, partitioning teams into DevOps, dev, and agentic workflows
- Evolved RAG stack from LangChain to Vespa to zero-ops Bedrock KB, improving retrieval and reducing overhead
- Continue as technical advisor on architecture and agentic-workflow direction

Chevron New Energies May 2024 – Aug 2024
Software Engineer Intern *Houston, TX*

- Implemented CI/CD workflows using Azure DevOps and K8s, enabling automated deployment and rollback
- Built RESTful APIs and data pipelines using Azure Data Factory, improving product efficiency by 40%

Projects

Kepler | *Rust, C++, Raft, LSM-tree, MVCC, Tokio* Mar. 2026 – Present

- Built a distributed, linearizable key-value store in Rust with from-scratch Raft consensus and LSM-tree engine
- Implemented Raft election, log replication, and membership changes for fault tolerance under node failure; storage: memtable, WAL, leveled SSTables
- Added MVCC transaction layer providing snapshot isolation and linearizable reads across multi-node Raft cluster
- Built single-tier compaction with full SSTable rewrite every 4 flushes; inline, writer-blocking execution

NOAA AUV Sonar Detection | *Python, C++, PyTorch, AWS Neuron SDK, SageMaker* April 2025 – July 2025

- Designed AWS Neuron SDK deployment path for onboard custom-silicon inference under hard real-time limits
- Trained CNN models on thermal sensor spectrograms to detect novel aquatic signatures for NOAA autonomous underwater vehicles
- Refined architecture and loss formulation to address class imbalance and sensor noise, improving model detection

Sentinel | *Next.js, FastAPI, Claude, pgvector, OpenTelemetry, Docker* Apr. 2026 – Present

- Built an AI SRE agent autonomously triaging incidents via tool-use over Prometheus, Tempo, and Loki telemetry
- Built pgvector RAG over runbooks and postmortems with prompt caching on frozen schemas; 100% eval accuracy
- Human-in-the-loop remediation gated behind approval; running live in SculptAI production triaging incidents

Skills & Honors

Languages/Tools: C, C++, Rust, Java, Python, Bash, Linux, TypeScript, JavaScript, Docker, Kubernetes, Jenkins, Buildkite, Groovy, Terraform, AWS, Azure, PostgreSQL, GraphQL

Systems & Reliability: Distributed consensus (Raft), concurrency, real-time/embedded inference, computer networks, systems security, CI/CD, observability (Datadog, OpenTelemetry)

Frameworks & Libraries: Spring, Flask, FastAPI, PyTorch, TensorFlow, Pandas, Node.js, Express, React, Next.js

Honors: Dean's List (All Semesters), Tech Lead at Northeastern Electronic Racing, PawHacks, json-io OSS Contributor